FreelanceSafe Lite

Milestone Escrow, Rule-Based Risk Scoring and Dispute Resolution for Freelancers

Complex Computing Problem – Semester Project Proposal

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The Problem

- Freelancers risk unpaid work; clients risk paying for poor or fake work. Trust is the main problem.
- Existing platforms charge high fees and resolve disputes slowly, often with little evidence-based reasoning.
- Fraud is varied: fake profiles, stolen payments, undelivered or poor work and false refund claims.
- No single right answer: security, fairness, speed and low friction pull in different directions.

Goal

Hold payments in milestone escrow, flag risky accounts with a simple rule-based score, and resolve disputes with transparent, evidence-based recommendations.

Proposed Solution: Five Interacting Components

1. Escrow Engine

Milestone contracts, simulated payments, release and refund rules

2. Risk Scoring

Rule-based score from account and payment signals

3. Reputation

Updates trust scores after each completed contract

4. Dispute Resolution

Collects evidence, recommends outcomes, supports appeals

5. Dashboard & Audit

Role-based views, reputation, alerts, audit log

Data flow: contract → escrow funded → work submitted → risk check → release or dispute → reputation update (Spring Boot layers: Controller → Service → Repository)

Why This Is a CCP

Depth of Knowledge

Secure state-machine design, role-based security, rule-based risk scoring

Conflicting Requirements

Security vs ease of use; fairness to both sides vs fast decisions

Depth of Analysis

Compare scoring rules and thresholds; no single obvious solution

Stakeholders

Freelancers, clients, platform admins, mediators, payment providers

Interdependence

Escrow state, fraud scores, evidence and reputation affect each other

Consequences

Wrong decisions cause financial loss, unfair disputes and data leaks

Trade-offs and Alternatives We Will Analyze

Conflicting requirements

- Strict fraud checks vs smooth onboarding for honest users
- Fast automatic dispute decisions vs fairness and human review
- Evidence collection vs privacy of work files

Alternatives compared

- Risk: simple threshold rule vs weighted score
- Disputes: admin-only decision vs weighted evidence score plus admin
- Metrics: false-block rate, missed-risk rate, dispute agreement, resolution time

Methodology, Tools and Evaluation

- Data: hand-made test scenarios for honest and risky contracts
- Stack: Java 17, Spring Boot, Spring Security, JPA/Hibernate, MySQL, Thymeleaf dashboard; simulated payments only
- Explainability: every risk flag and dispute recommendation lists its evidence
- Validation: JUnit/Mockito tests plus scenarios (honest deal, non-payment, fake refund claim, invalid state change)
- Baseline: compare against manual dispute handling without scoring

Timeline and Expected Outcomes

Wk 1-3
Requirements, data, literature

Wk 4-6
Java architecture, escrow engine, test data

Wk 7-9
Risk score, dispute, reputation

Wk 10-12
Dashboard, integration, testing

Wk 13-14
Evaluation, report, demo

- Working prototype with escrow flow, dashboard and explanations
- Test results showing invalid payment transitions are blocked
- Documented trade-off analysis and ethical/privacy safeguards